

ENTERPRISE INFRASTRUCTURE PLAN

ITEP 414 - System Administration and Maintenance

ABC STARTUP SOLUTIONS

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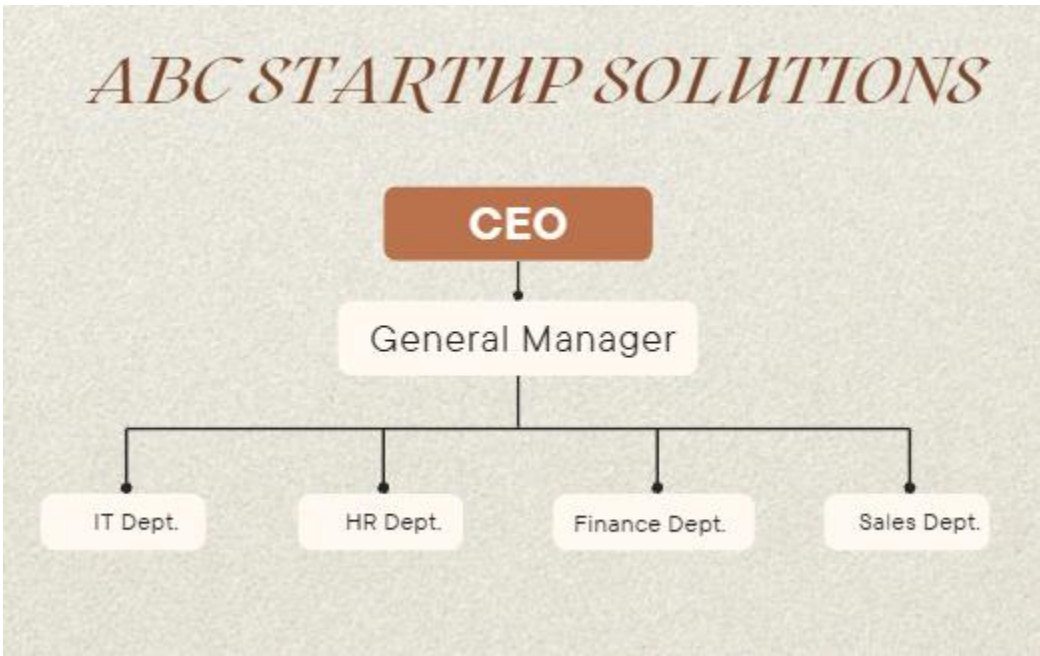
Program: Bachelor of Science in Information Technology

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PART 1: Company Profile

- **Company Name:** ABC Startup Solutions
- **Nature of Business:** Software Development Company
- **Company Vision:** To become a premier software development partner in Southeast Asia, delivering scalable, secure, and modern applications that power digital transformation.
- **Company Mission:** To design and deploy high-performance custom software solutions using robust IT infrastructure, modern continuous integration workflows, and industry-standard security frameworks.
- **Office Location:** 5th Floor, Cybertech Tower, Santa Rosa City, Laguna, Philippines
- **Organizational Structure:**



- **Employee Distribution**

Department	Number of Employees	Core Operational Function
Information Technology	5	Software engineering, dev environments, system admin, network security, and internal IT support.
Human Resources	4	Talent acquisition, employee onboarding, performance management,

		and HR records management.
Finance	5	Corporate accounting, payroll processing, budget allocation, and billing operations.
Sales & Marketing	6	Client outreach, product demonstrations, contract negotiations, and digital marketing.

PART 2: Enterprise Hardware Inventory

Asset ID	Hardware Item	Quantity	Department	Purpose & Technical Justification
HW-DESK-01	Workstation PC (Core i7 / Ryzen 7, 32GB RAM, 1TB NVMe SSD)	5	IT Department	Needed for local application builds, running Docker containers, and virtual machines.
HW-MON-01	27" Dual-Monitor Setup	10	IT Department	Expanded display space to improve coding productivity and debugging efficiency.
HW-LAP-01	Business Laptop (Core i5 / Ryzen 5, 16GB RAM, 512GB SSD)	15	HR (4), Finance (5), Sales (6)	Provides mobility for client meetings, presentations, and general office productivity.
HW-MON-02	24" Single Monitor	15	HR (4), Finance (5), Sales (6)	Ergonomic desktop extensions for laptops when employees work at their office desks.
HW-SRV-01	Rackmount Dell PowerEdge Server (64GB RAM, 4TB RAID Array)	1	Server Room (IT)	Host server for local domain controller, Active Directory, internal file sharing, and Git repositories.
HW-RTR-01	Enterprise Router (MikroTik / Cisco)	1	Server Room (IT)	Handles WAN connectivity, inter-VLAN routing, network address translation (NAT), and firewalling.
HW-SW-01	48-Port Managed Gigabit PoE+ Switch	1	Server Room (IT)	Central switch providing wired internet access to workstations, servers, access points, and printers.

HW-WAP-01	Wi-Fi 6 Dual-Band Access Point	2	Office Ceiling Layout	Delivers high-speed wireless coverage for laptops, mobile devices, and guests.
HW-NAS-01	Network Attached Storage (Synology 4-Bay, 16TB Raw Storage)	1	Server Room (IT)	Centralized local backup target for file archives, project assets, and database backups.
HW-EXT-01	4TB External Hard Drive	2	IT Department	Physical cold storage drives used for offsite physical backup rotation.
HW-UPS-01	3000VA Smart-UPS (Rackmount)	1	Server Room (IT)	Battery backup protecting servers, core switch, router, and NAS against unexpected power outages.
HW-UPS-02	650VA Desktop UPS	5	IT Department	Surge protection and clean shutdown time for developers' workstation PCs.
HW-PRN-01	Enterprise Multifunction Laser Printer	1	Common Office Area	Shared printer/scanner for HR contracts, financial receipts, and legal documents.

PART 3: Enterprise Software Inventory

Software	Version	License Type	Purpose & Technical Justification	Software
Windows 11 Pro	22H2 / Latest	Volume Commercial License	Business OS providing BitLocker disk encryption, remote desktop, and Active Directory Domain Join.	Windows 11 Pro
Ubuntu Server	22.04 LTS	Open Source (Free)	Production server OS hosting internal database engines, web services, and automation scripts.	Ubuntu Server
Microsoft 365	Business Standard	Monthly Subscription	Primary communication suite (Outlook, Teams, Word, Excel, OneDrive) across all 20 employees.	Microsoft 365
VS Code	Latest Stable	Open Source (Free)	Main source code editor for software development projects.	VS Code

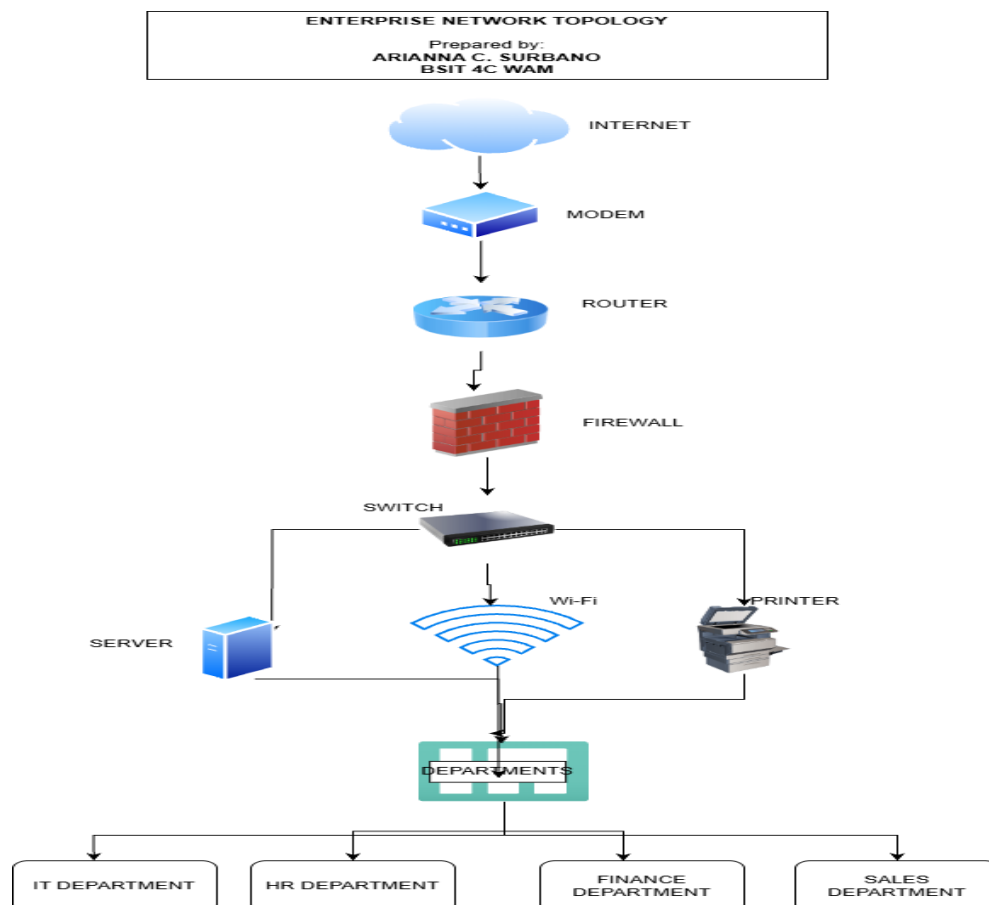
Git	Latest Version	Open Source (Free)	Command-line distributed version control system for software development tracking.	Git
GitHub Desktop	Latest Version	Open Source (Free)	Graphic interface facilitating Git workflow management for developers.	GitHub Desktop
VirtualBox	7.x	Open Source (Free)	Desktop virtualization software for testing operating systems and isolated application builds.	VirtualBox
Google Chrome	Latest Version	Freeware	Standard secure enterprise web browser for SaaS access and web debugging.	Google Chrome
Microsoft Defender for Endpoint	Enterprise	Enterprise Subscription	Centralized Endpoint Detection and Response (EDR), antimalware, and threat prevention.	Microsoft Defender for Endpoint
AnyDesk	Professional	Commercial Subscription	IT support utility enabling remote desktop access for troubleshooting endpoint issues.	AnyDesk
7-Zip	Latest Version	Open Source (Free)	Secure archive utility for file compression and extraction.	7-Zip

PART 4: Enterprise Network Inventory

Hardware Component	Brand / Specifications	Quantity	Functionality
ISP Modem	Fiber Optic Terminal	1	Terminates incoming ISP fiber connections and converts them to Ethernet WAN traffic.
Enterprise Router	Cisco / MikroTik Gigabit Router	1	Handles WAN IP routing, bandwidth management, dynamic host configuration (DHCP), and VPN access.

Hardware Firewall	Fortinet FortiGate / pfSense	1	Perimeter firewall for stateful packet inspection, Intrusion Prevention System (IPS), and web filtering.
Managed Switch	Cisco 48-Port PoE+ Managed Switch	1	Managed core switch handling VLAN tagging (802.1Q) and powering Wireless Access Points.
Access Points	Ubiquiti UniFi 6 Pro	2	Enterprise Wi-Fi access points deployed across the floor for full wireless coverage.
Patch Panel	48-Port Cat6 Patch Panel	1	Terminates structured horizontal cable runs from wall drops to the network rack.
CAT6 Ethernet Cables	Cat6 UTP Cable Box (1000 ft)	2 Boxes	High-speed network cables connecting wall plates to switches and workstations.
RJ45 Connectors	Cat6 Pass-Through Connectors	200 Pcs	Connectors used for crimping custom patch cables and network drops.

PART 5: Enterprise Network Diagram



PART 6: System Administration Roles

1. Helpdesk Technician

- **Responsibilities:** Serves as the first point of contact for technical support, logs and manages support tickets, builds and deploys end-user workstations, and troubleshoots basic hardware/software faults.
- **Skills:** Operating system diagnostics (Windows/macOS), hardware repair, basic networking troubleshooting, active listening, and customer service skills.
- **Common Tools:** Ticketing software (Jira Service Desk, Freshdesk), Active Directory Users & Computers, AnyDesk, and Remote Desktop Protocol (RDP).
- **Certifications:** CompTIA A+, Microsoft Certified: Modern Desktop Administrator Associate.

2. Network Administrator

- **Responsibilities:** Configures, manages, and maintains physical and virtual network infrastructure; manages switches, routers, firewalls, and Wi-Fi infrastructure; monitors network performance and security policies.
- **Skills:** Routing and switching protocols (OSPF, BGP, VLAN tagging 802.1Q), network security, firewall rule administration, and IP address management (IPv4/IPv6).
- **Common Tools:** Wireshark, Cisco Packet Tracer, PuTTY/Solar-PUTTY, SolarWinds, and Nmap.
- **Certifications:** Cisco Certified Network Associate (CCNA), CompTIA Network+.

3. Linux System Administrator

- **Responsibilities:** Installs, configures, and maintains Linux servers; manages file permissions, system updates, server security hardening, shell scripting, and application container environments.
- **Skills:** Command-line interface (CLI) proficiency, Bash and Python scripting, systemd service management, storage management (LVM), and UFW/iptables firewall configuration.
- **Common Tools:** OpenSSH, Ansible, Docker, Vim/Nano, htop, and journalctl.
- **Certifications:** Red Hat Certified System Administrator (RHCSA), CompTIA Linux+, LPI Linux Essentials.

4. Cloud Administrator

- **Responsibilities:** Provisions, monitors, and optimizes cloud computing assets (IaaS/PaaS/SaaS); manages Identity and Access Management (IAM) permissions; sets up cloud backup target pipelines and manages cloud spend.
- **Skills:** Cloud security posture, virtual private network setup (VPC/VNet), Infrastructure as Code (IaC), cost optimization, and automated cloud monitoring.

- **Common Tools:** AWS Management Console, Microsoft Azure Portal, Terraform, AWS CLI, and CloudWatch.
- **Certifications:** AWS Certified Solutions Architect – Associate, Microsoft Certified: Azure Administrator Associate.

Organizational Collaboration

In an enterprise technology infrastructure, these four roles operate as an interconnected unit. The **Helpdesk Technician** triages end-user requests and escalates infrastructure-level network drops or server outages. The **Network Administrator** maintains stable network connectivity and security boundaries, allowing the **Linux System Administrator** to deploy servers, web services, and repositories securely. Simultaneously, the **Cloud Administrator** bridges on-premises servers with cloud backups and hybrid tools. Together, they safeguard operational uptime, data integrity, and business continuity.

PART 7: Infrastructure Recommendations

- **Internet Provider (ISP):**
 - *Recommendation:* Primary Dedicated Enterprise Fiber Line (500 Mbps) backed up by a secondary connection (100 Mbps) from an alternate provider configured for dynamic WAN failover.
 - *Reasoning:* Software development requires reliable uplink/downlink bandwidth. A dedicated leased line guarantees 99.9% Service Level Agreements (SLA), preventing costly work interruptions.
- **Server Specifications:**
 - *Recommendation:* Enterprise Rackmount Dell PowerEdge R450 Server with an Intel Xeon Silver Processor, 64GB DDR4 ECC Memory, and 4x 2TB Enterprise SAS Hard Drives configured in RAID 10.
 - *Reasoning:* ECC RAM prevents random data corruption, while RAID 10 delivers superior write/read speeds and high fault tolerance (allowing drive loss without data destruction).
- **Backup Strategy:**
 - *Recommendation:* Implementation of the **3-2-1 Backup Rule**.
 - *Reasoning:* Maintain 3 copies of business data (1 Primary production data, 1 Local target on Synology NAS, and 1 Encrypted Offsite Cloud Backup) across 2 different media types, ensuring 1 copy is physically offsite.
- **Security Policies:**
 - *Antivirus/EDR:* Centralized deployment of Microsoft Defender for Endpoint across all computers and servers for real-time telemetry, threat isolation, and ransomware detection.
 - *Password Policy:* Enforce complex passwords (minimum 12 characters, requiring uppercase, lowercase, numbers, and special characters), multi-factor authentication (MFA) across all M365 and VPN logins, 90-day password cycling, and automatic screen locks after 5 minutes of inactivity.

- **Expansion Plan:**

- *Recommendation:* Structured network switch provision and dynamic subnets.
- *Reasoning:* The 48-port core switch utilizes ~25 ports at launch, leaving capacity for 100% workforce expansion on the floor without requiring additional rack switches.

PART 8: Personal Reflection

Through completing this project, I gained practical insight into the holistic planning required to build an enterprise IT ecosystem from scratch. I realized that system administration requires balancing operational requirements, security standards, financial budgets, and scalable expansion plans rather than merely repairing hardware issues.

The most challenging task was designing the enterprise network topology diagram and structuring logical VLAN segmentations. Ensuring that sensitive traffic from Finance and HR was completely isolated from guest Wi-Fi users while allowing secure routing to shared network resources required careful subnet planning and firewall positioning.

Planning prior to deployment is critical because purchasing equipment without architectural blueprints leads to severe operational bottlenecks, security vulnerabilities, incompatible hardware purchases, and costly downtime. Proper planning ensures that systems remain maintainable, secure, and ready to scale alongside the company.

This project enhances my growth as a System Administrator by strengthening my technical documentation skills, network design competency, hardware budgeting ability, and understanding of security frameworks—skills essential for delivering enterprise-level IT support.